

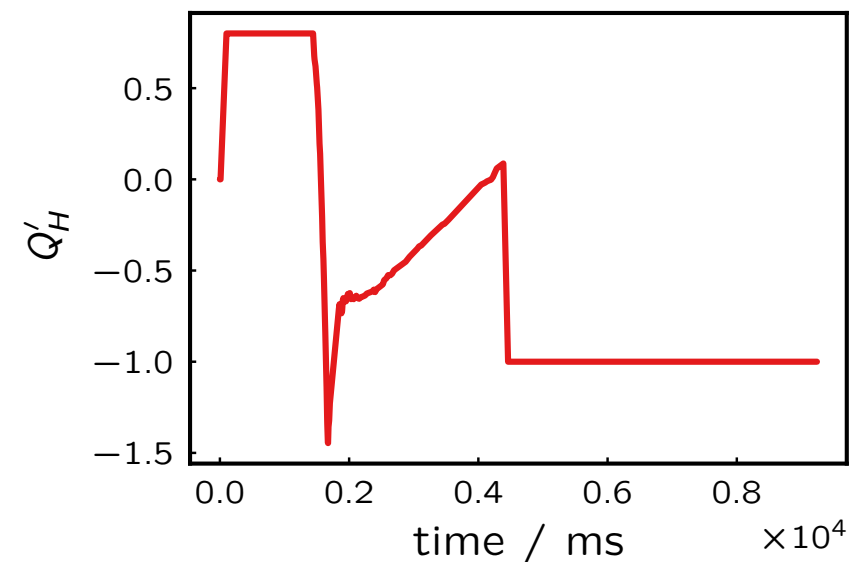
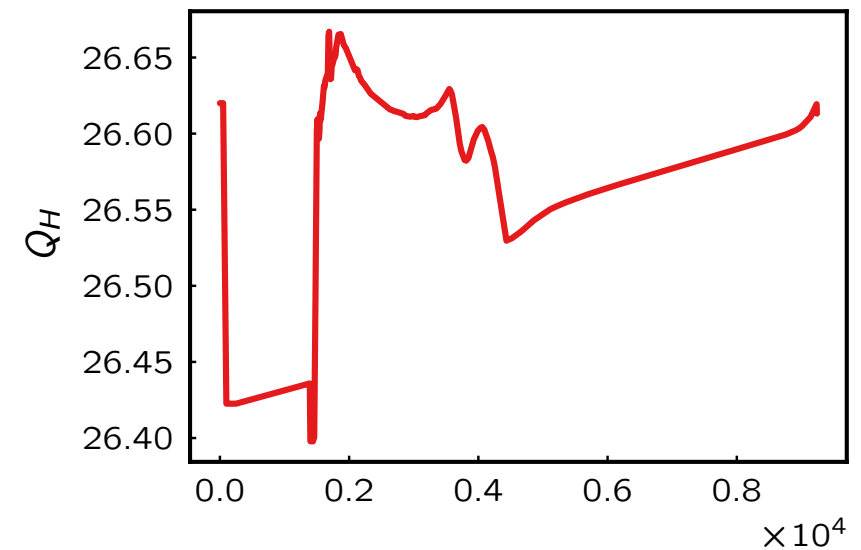
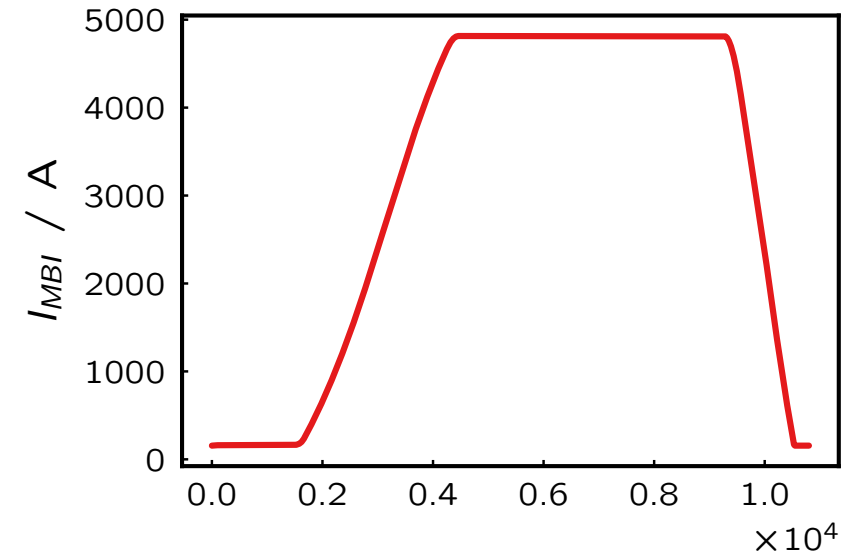
# Constant Optics Slow Extraction (COSE)

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J.P. Prieto, L.S. Stoel

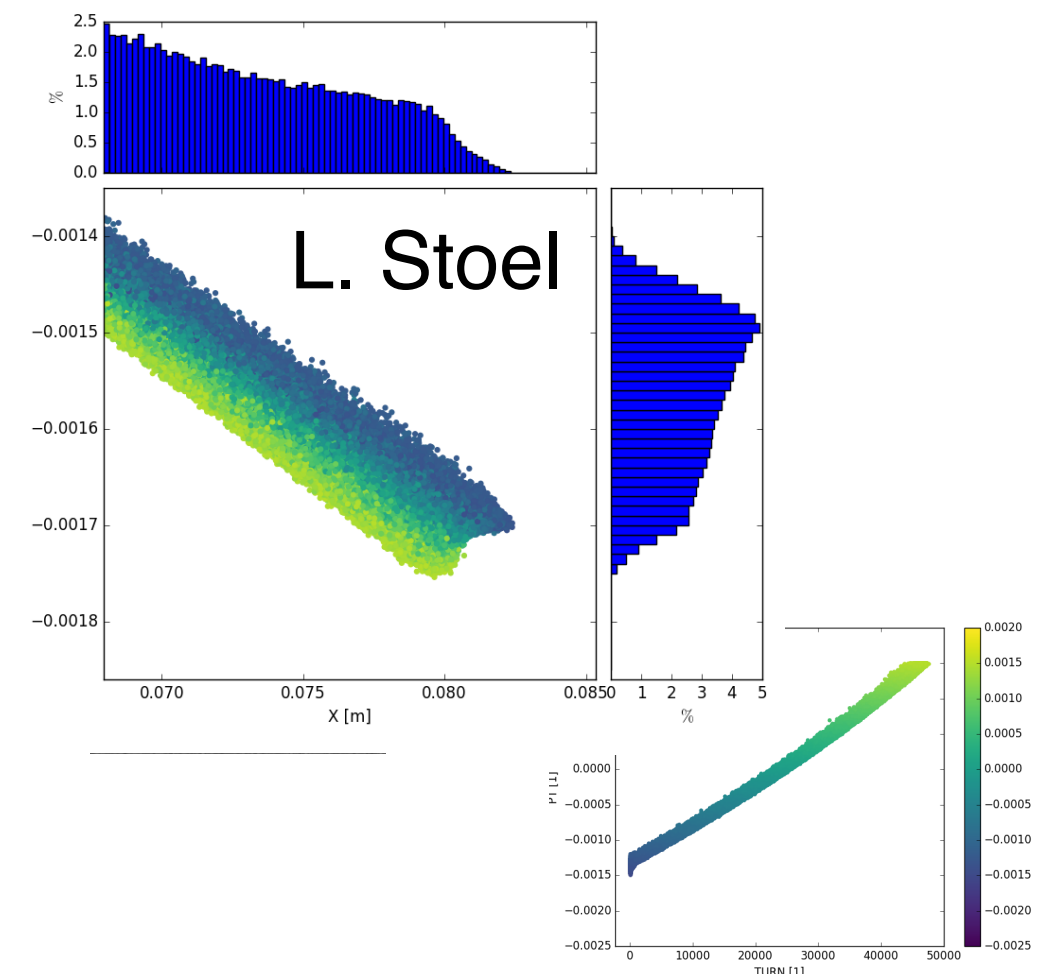
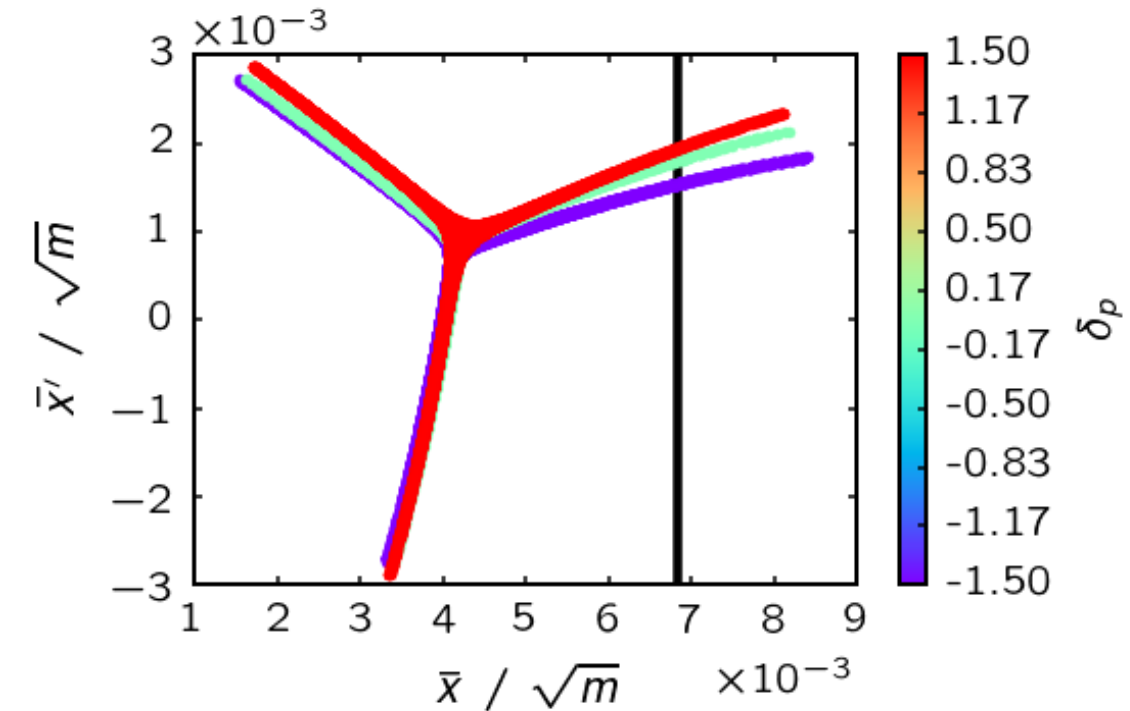
# Today's slow extraction

- Today's SE is done with the following steps:
  - ↳ Beam accelerated at 400 GeV/c
  - ↳ Tunes set up at 26.62/26.58
  - ↳ Extraction bump ON
  - ↳ Chromaticity set to -1 (trim settings)
  - ↳ Harmonic sextupoles ON
  - ↳ RF off after end of RF gymnastic (start of flat top)
  - ↳ QF and QD swept to change the tune from 26.62 to 26.70 (theoretically)



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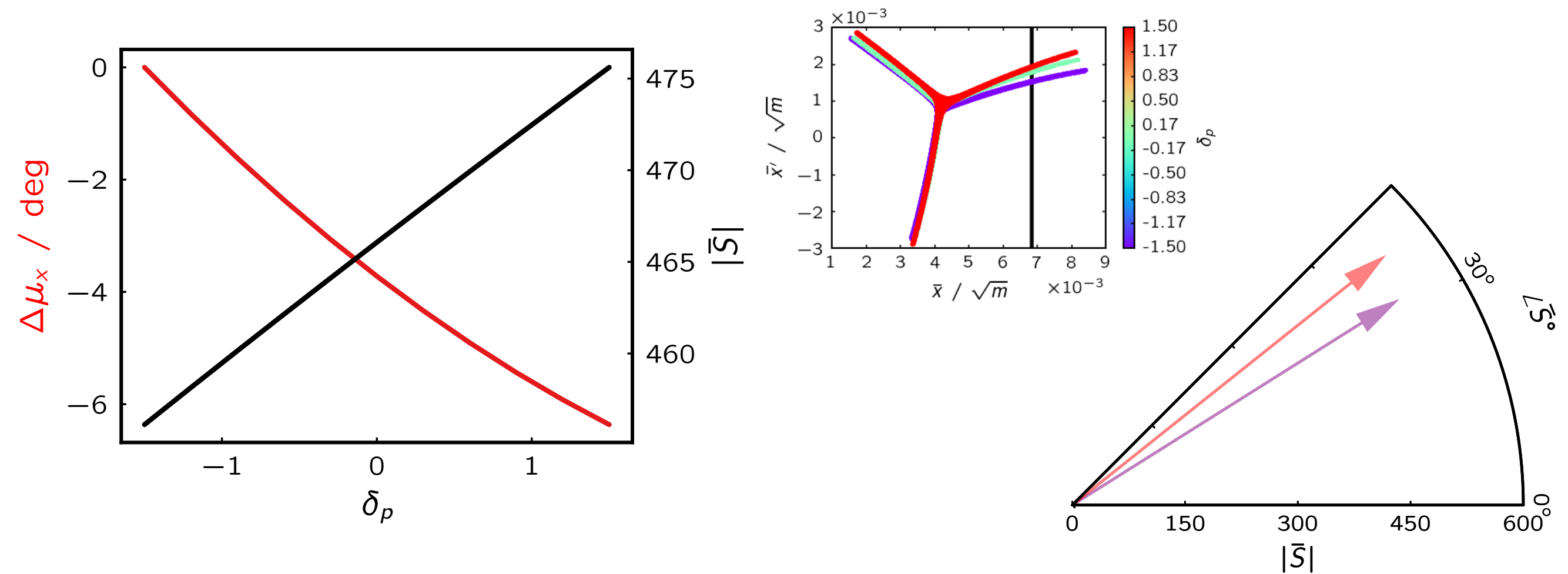


# Today's slow extraction

- With the virtual sextupole approximation, the effect of the quadrupoles sweeping is clear

$$|\bar{S}| = \sqrt{\sum_n (S_n \cos(3\mu_{x,n}))^2 + \sum_n (S_n \sin(3\mu_{x,n}))^2} \quad \angle \bar{S} = \frac{1}{3} \tan^{-1} \left( \frac{\sum_n S_n \sin(3\mu_{x,n})}{\sum_n S_n \cos(3\mu_{x,n})} \right)$$

- As the optics is NOT constant, the phase advance between the virtual sextupole and the ZS entrance changes in time ( $\Rightarrow$  with the momentum of the particles extracted)



→ Constant Optics Slow Extraction => COSE

- ↳ Today we “only” change the QF and QD to sweep the tune in order to extract particles with different momenta as a function of time
- ↳ This means that the rest of the machine (bends, sextupoles, extraction elements) are kept constant during the spill
- ↳ This causes the effect shown before - the optics of the machine is actually changing (e.g. natural chromaticity...) during the spill
- ↳ Hence the idea to “scale” the whole machine with the extracted particles momentum => **Constant optics over the spill**

→ How does it work?

↳ The Qh is set up to be at (theoretically) 26.62

↳ The beam energy is changed (radial trim) to  $+1.5e-3$  (assuming momentum spread  $\pm 1.5e-3$ )

‣ In this way, particles that have the lowest momentum are centred in the vacuum chambers => their mean CO should be zero...

↳ Now, ideally, one should change by the same relative amount each of the active elements current (keeping the normalised strength k constant) to follow the extracted beam energy...

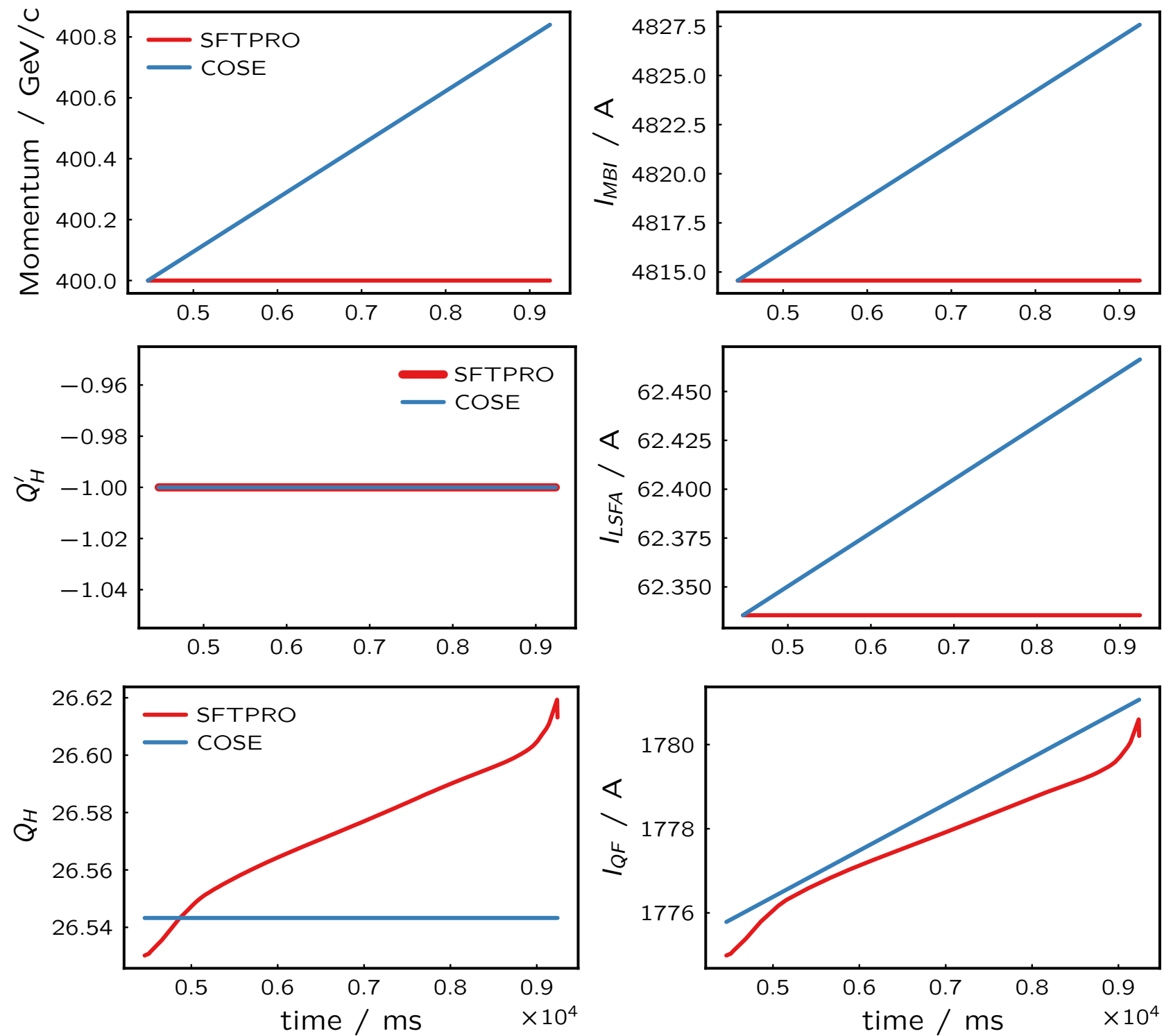
→ Thanks to LSA, this can be done trimming “just” one parameter: momentum



**...way to big to fit in one slide!!!**

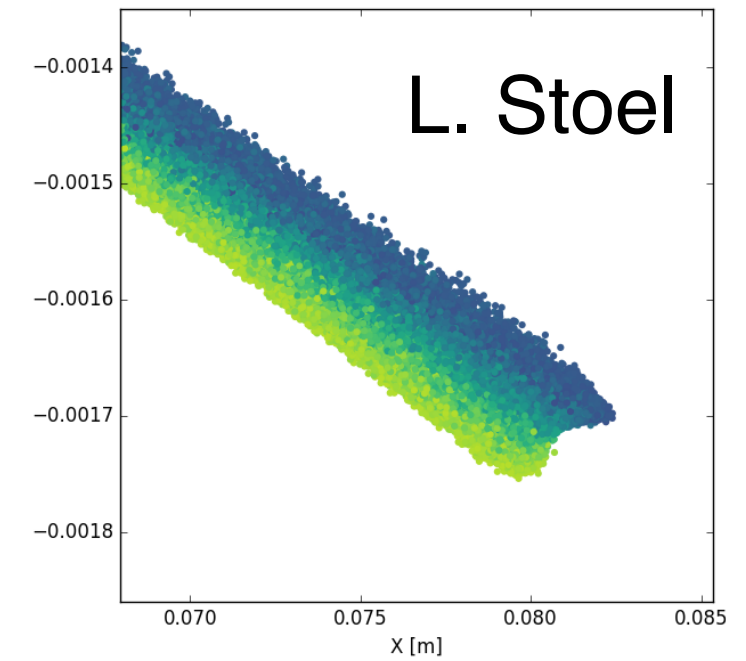
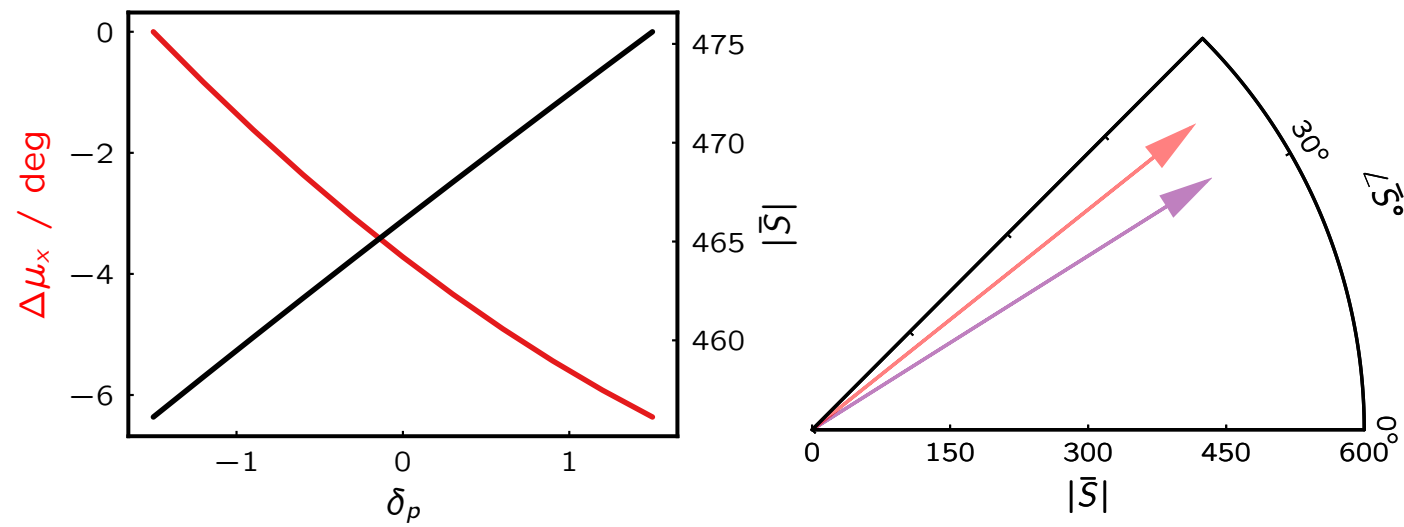
→ Constant Optics Slow Extraction => COSE

↳ Looking at the LSA function for the main SPS elements:



→ Constant Optics Slow Extraction => COSE

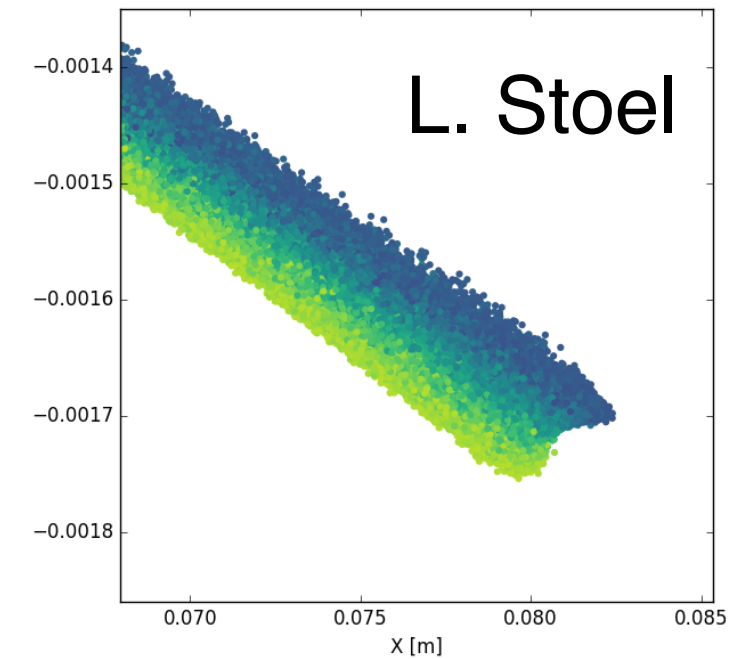
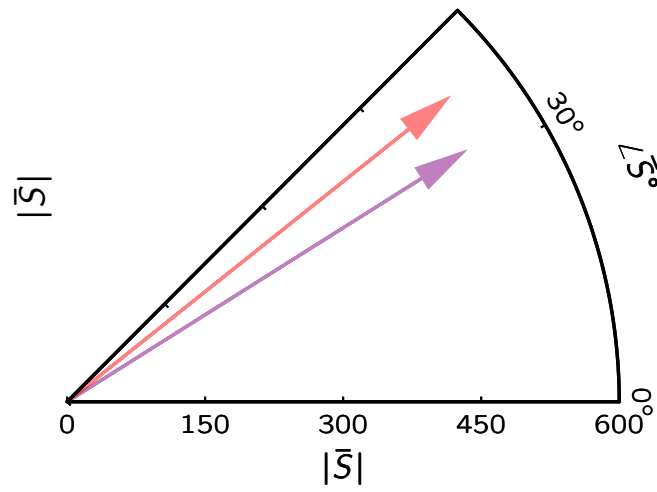
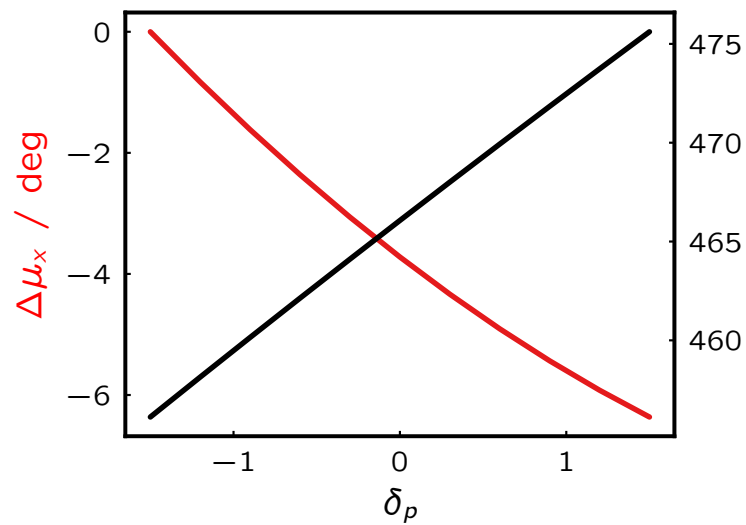
↳ Looking at the extraction separatrix:



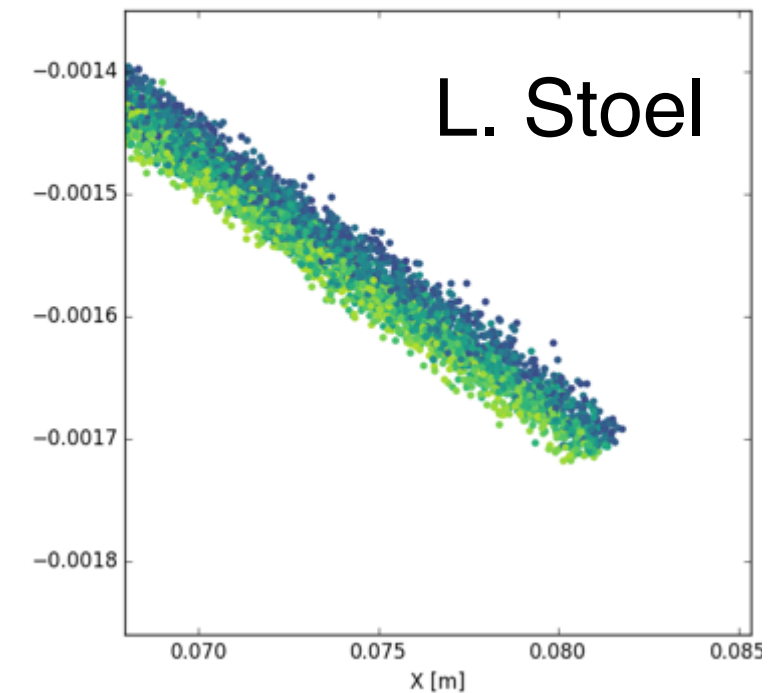
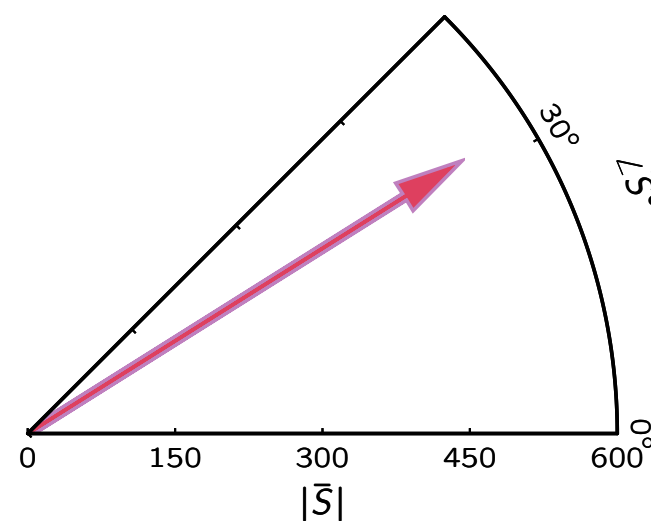
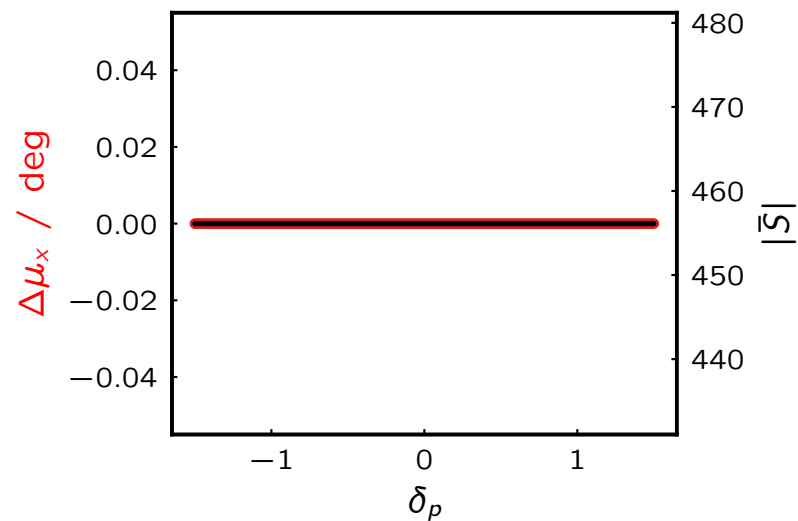
# COSE

→ Constant Optics Slow Extraction => COSE

↳ Looking at the extraction separatrix:



COSE



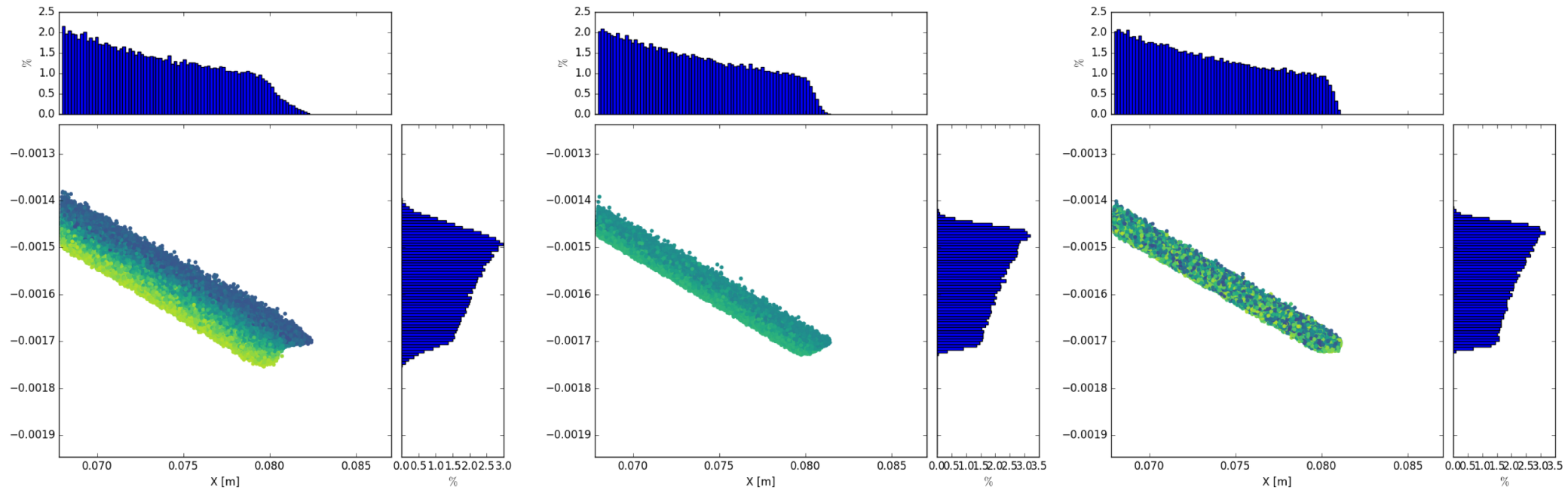
# COSE during MDs

## Beam on grids

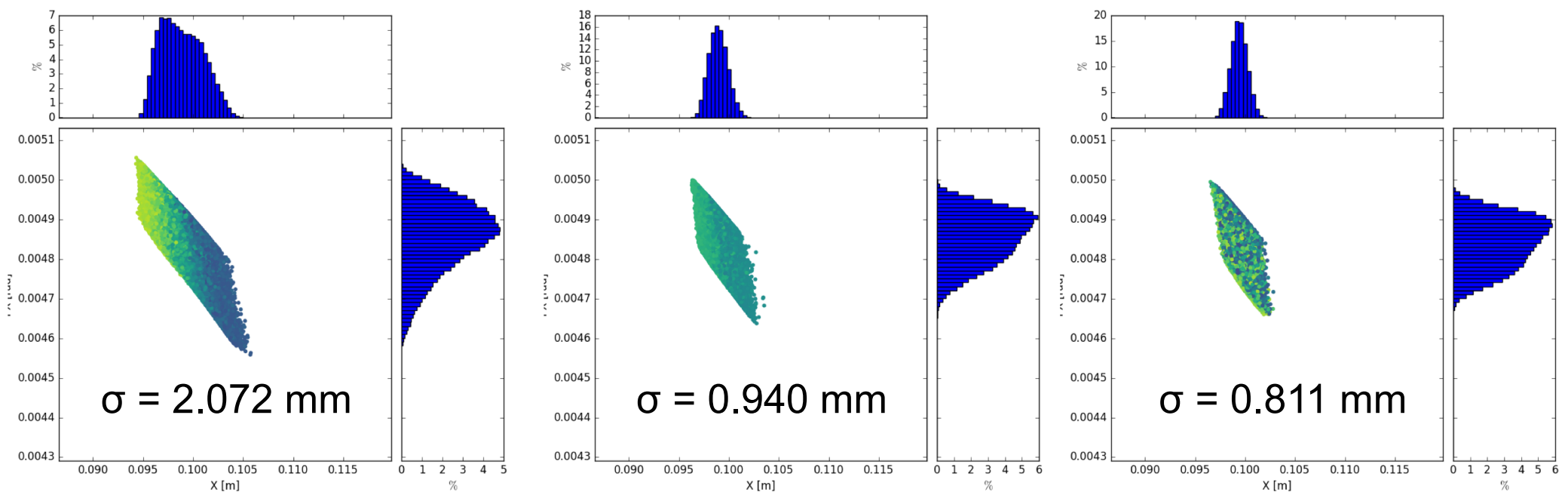
- Left: Nominal
- Middle: Nominal but dpp 0.5 permille
- Right: scaled optics dpp 1.5 permille

L. Stoel

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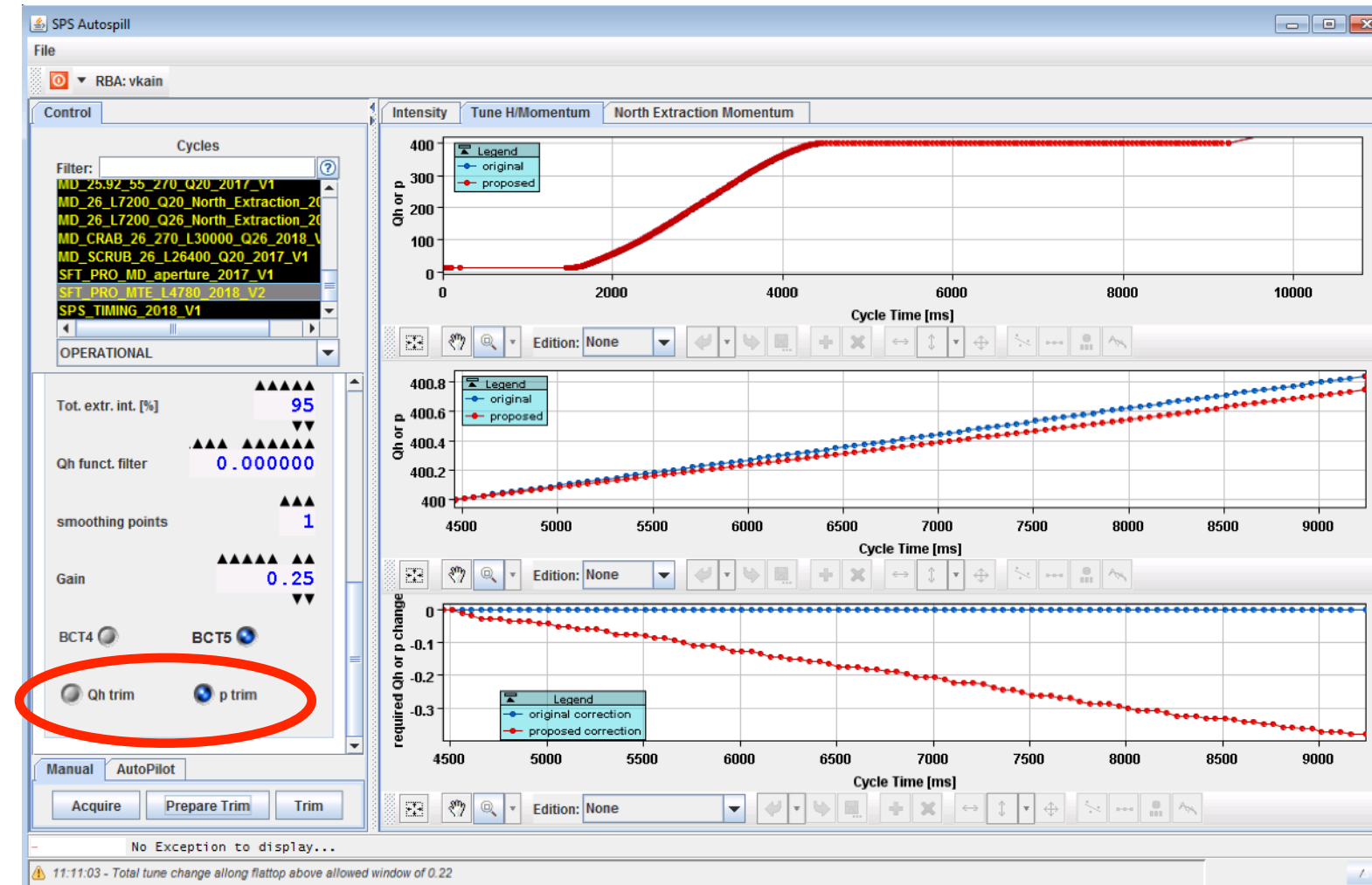
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# COSE for FT operation

→ What is still missing to make COSE operational?

- ↳ Auto-spill to work on momentum - the application is ready, it only needs time for debugging and to be tried out
- ↳ Compare chromaticity measurements of the two methodology
- ↳ Test with nominal intensity and compare with normal SE
  - Are the losses improved ( $Dp/p$  larger)?
- ↳ Test transport through TT20 to the targets



→ This means about 2 MD slots to complete (max)

- With today's way of driving the slow extraction quadrupoles, the optics of the machine changes during the spill
  - ↳ This causes the rotation of the extraction separatrix during the spill, translating in a large angular footprint at the ZS extraction
- The idea then to scale the whole machine (active elements) to follow the energy of the extracted particles
  - ↳ This led to the concept of the so-called Constant Optics Slow Extraction => COSE
- Technically, this can be done exploiting the LSA high level parameters, i.e. MOMENTUM
  - ↳ Currents of all active elements varied to follow the beam energy
- A few measurements still missing before deploying it operationally